The 5 Pro Scalps — can the tool build them?

Source: `trades_.pdf` (RubberBand, Second Chance, Back$ide, HitchHiker, Fashionably Late). This document (1) states each scalp in plain logical language, (2) shows the exact builder conditions that express it, and (3) notes what the engine approximates.

Verdict: all 5 entries and their core exits build with the primitives that already exist — no new primitives were added. Every strategy here is constructed as real strategy JSON and fires through the full `evaluate()` path in the regression suite (`chart/tests/logic_audit17.py`, part 18). Scale-outs (thirds / halves / waves) are modelled as exit legs (see below); entry-frozen measured-move targets are approximated with an anchored line.

Long side is shown throughout; every scalp inverts for shorts (swap `gt` ↔ `lt`, `cross_above` ↔ `cross_below`, `bounce_up` ↔ `bounce_down`, `today_low` ↔ `today_high`, and set Side = Short).

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1. RubberBand — mean-reversion snapback (Side: Long)

Idea. Stock over-extends DOWN, selling gets sloppy, then snaps back.

Logical rules

- Entry: a **green candle** whose **high clears the highs of the prior 2 candles** (the "double-bar break"), while the move is extended and volume is climbing.
- Stop: ~$0.02 below the **low of the day**.
- Exit: scale out — 1/3 at 1:1, 1/3 at 2:1, final 1/3 into **VWAP**.

Builder (ENTRY, ALL match)

left	op	right
Expr `day_open − today_low`	`>`	Expr `3 × atr_daily`
Price `close`	`>`	Price `open`
Price `high`	`>`	`extremes.highest` len 2, source high, **[1] ago**

The first row is the **defining premise** and ships in the seed: the day must be extended DOWN by >3 ATR (open to low-of-day) — without it the double-bar break fires in any uptrend, the opposite of a snapback fade.

The PDF's **"RVOL > 5"** is the SCREENER's daily relative volume — a stock-**selection** filter, not an intraday bar condition. It is already handled by trading only the R1 universe (these ARE the high-RVOL in-play names) and is filterable in the backtest by the `rvol` column, so it is not an entry rule. Time-of-day `hhmm` 1000–1330 is an optional filter you can add.

Stop: SL = anchored to `today_low`, small % beyond (the $0.02 becomes a tiny %; the exact 2-cent offset is a bracket detail).

Exit that lives in the engine: `close crosses ▲ vwap.session` (the final third into VWAP). → **The 1:1 and 2:1 partials are trading-tool bracket legs.**

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2. Second Chance — breakout retest, old resistance becomes support (Long)

Logical rules

- Break a resistance level, **retest** it from above, then a candle **closes above the prior candle** (buyers confirm).
- Stop: ~$0.02 below the low of the turn candle.
- Exit: half at the pullback high, trail the rest under the **9 EMA**.

Builder (ENTRY, IN SEQUENCE / THEN, window ≈ 6 bars)

step	left	op	right
1 break	Price `close`	`crosses ▲`	`structure.pivot_high` **(hold)** — the range top, held as a fixed level
2 retest	Price `low`	`≤`	the **same** held `pivot_high`
3 attack	Price `close`	`>`	Price `close` **[1] ago**

The resistance is not a single arbitrary level but the **top of the range the stock broke** — a real swing high (`structure.pivot_high`), **held** forward (`"hold": true`) so it stays fixed through the retest. A rolling `highest(N)` can't be used: after the break it jumps to the breakout high and smears the retest. (The old seed used the first-4-min opening-range high — usually a tiny, meaningless level; that produced the junk breaks in backtest #75.) `pm_high` / `prev_day_high` are alternative fixed levels if you want a specific one.

Stop: the PDF says ".02 below the low of the **turn candle**" → anchored to `extremes.lowest` len 3, source low (the recent retest low), not the day low. **Exit in the engine:** `close crosses ▼ ema(9)` (the trail). → **The "sell half at the pullback high" leg is a trading-tool bracket.**

The level is your choice of any real level primitive — that is exactly the "old resistance becomes support" idea made concrete.

3. Back$ide — reversal back to VWAP (Long)

Logical rules

- After a low, price builds **higher lows above a rising 9 EMA**, then **breaks the recent range**; ride it to **VWAP**.
- Stop: ~$0.02 below the most recent higher low. One attempt.
- Exit: **entire** position at VWAP.

Builder (ENTRY, ALL match) — all 5 conditions ship in the seed

left	op	right
Price `close`	`>`	`ema` len 9
`ema` len 9	`rising`	(upsloping 9EMA)
`extremes.lowest` len 3, source low	`rising`	**the "higher low"**
Price `close`	`>`	Expr `(today_low + vwap.session)/2` (range above the midpoint)
Price `close`	`crosses ▲`	`extremes.highest` len 5, source high, **[1] ago** (the "higher high"/range break)

Stop: PDF says ".02 below the **most recent higher low**" → anchored to `extremes.lowest` len 5, source low (a recent swing low), not the day low. **Exit in the engine:** `close crosses ▲ vwap.session` — **fully expressible, single target. This scalp maps 1:1.**

4. HitchHiker — consolidation breakout after an opening drive (Long)

Logical rules

- Drive up, then a tight **consolidation** (low in the upper 1/3 of the day range), then **break the consolidation high on +30% volume**.
- Stop: ~$0.02 below the consolidation low.
- Exit: waves — 1/2 into the first wave, 1/2 into the second.

Builder (ENTRY, ALL match)

left	op	right
Price `close`	`crosses ▲`	`extremes.highest` len 6, source high, **[1] ago**
Price `close`	`>`	Expr: `today_low + (2/3)×(today_high − today_low)`
Price `volume`	`>`	Expr: `1.3 × volume [1] ago`

Optional: consolidation above `pm_high` / `prev_day_high`; setup time `hhmm < 959`.

Stop: anchored to `extremes.lowest` len 6, source low (the consolidation low). **Exit in the engine:** `close crosses ▼ ema(9)` as a wave proxy. → **The precise two-wave scale-out is a trading-tool concern.**

Note: "consolidation" is approximated as *break of the rolling N-bar high while price sits in the upper third of the day's range* — a faithful, mechanical stand-in for the visual pattern.

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5. Fashionably Late — 9 EMA crosses VWAP (Long)

Logical rules

- Enter when an **upsloping 9 EMA crosses above a flat-to-down VWAP**.
- Stop: a **measured move** — 1/3 of the VWAP-to-LoD distance.
- Exit: 1 measured move above the cross.

Builder (ENTRY, ALL match) — all 3 conditions ship in the seed

left	op	right
`ema` len 9	`crosses ▲`	`vwap.session`
`ema` len 9	`rising`	**cons≥ 0** (see lesson below)
`vwap.session`	`≤`	`vwap.session` **[5] ago** (VWAP flat-to-down, per the PDF)

Stop: SL anchored to an Expr `vwap − (2/3)×(vwap − today_low)` (1/3 of the way down from VWAP to the LoD). **Target:** the measured move (cross + (cross − LoD)) is *frozen at entry*; on the chart it is approximated with a VWAP-anchored Expr.

Coverage summary

Scalp	Entry logic	Engine exit	Trading-tool part
RubberBand	✅ double-bar break + filters	✅ VWAP	1:1 & 2:1 partials
Second Chance	✅ break→retest→attack (THEN)	✅ 9EMA trail	sell-half leg
Back$ide	✅ HL + rising 9EMA + range break	✅ VWAP (1:1)	—
HitchHiker	✅ range break + upper-⅓ + volume	✅ 9EMA proxy	2-wave scale-out
Fashionably Late	✅ 9EMA×VWAP cross	✅ measured-move approx	frozen MM target

Two lessons this validation surfaced (real, not test artifacts)

1. `cross_above` + `rising`**: use consistency 0 at the cross.** At the exact bar a fast MA crosses up through a slow line, the fast MA has *just* turned — it is higher than N bars ago (direction up) but has not been *consistently* rising for N bars. A strict `cons≥` run only passes a bar or two later and would miss the cross. Express "upsloping" as pure direction (`cons≥ 0`), or just let the cross imply the upslope.
2. **Warm-up before the signal.** `ema(9)` and `vwap.session` both anchor at 09:30; an EMA-cross-VWAP is only meaningful after ~9+ bars. In real data these scalps fire mid-morning (10:00–10:45), long past warm-up, so this is never a problem live — but a preview window must include the bars before the cross (the engine already extends the fetch for warm-up and slices the display to your window).

Scale-out exits (now modeled — the seeds carry them)

The engine models partial exits: `risk.targets` = ordered legs `{fraction, r_multiple}` (or a fixed `tp`), the runner rides to SL/exit/eod, and the return is size-weighted. The seeds ship their PDF exits:

- **RubberBand** — ⅓ at 1R, ⅓ at 2R, final ⅓ (runner) into VWAP.
- **Second Chance** — ½ at ~2R, trail the other ½ under the 9-EMA.
- **HitchHiker** — ½ at 1R (wave 1), trail ½ under the 9-EMA (wave 2).
- **Back$ide / Fashionably Late** — single exit (the PDF has no scale-out).

The chart draws a teal T1/T2 (…%) tick at each partial; the report shows how many partials banked.